Problem Set 3

Applied Stats/Quant Methods 1

Due: November 13, 2025

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in `R`, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday November 13, 2025. No late assignments will be accepted.

In this problem set, you will run several regressions and create an add variable plot (see the lecture slides) in `R` using the `incumbents_subset.csv` dataset. Include all of your code.

Question 1

We are interested in knowing how the difference in campaign spending between incumbent and challenger affects the incumbent's vote share.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `difflog`.

```
lm(voteshare ~ difflog, inc.sub)
```

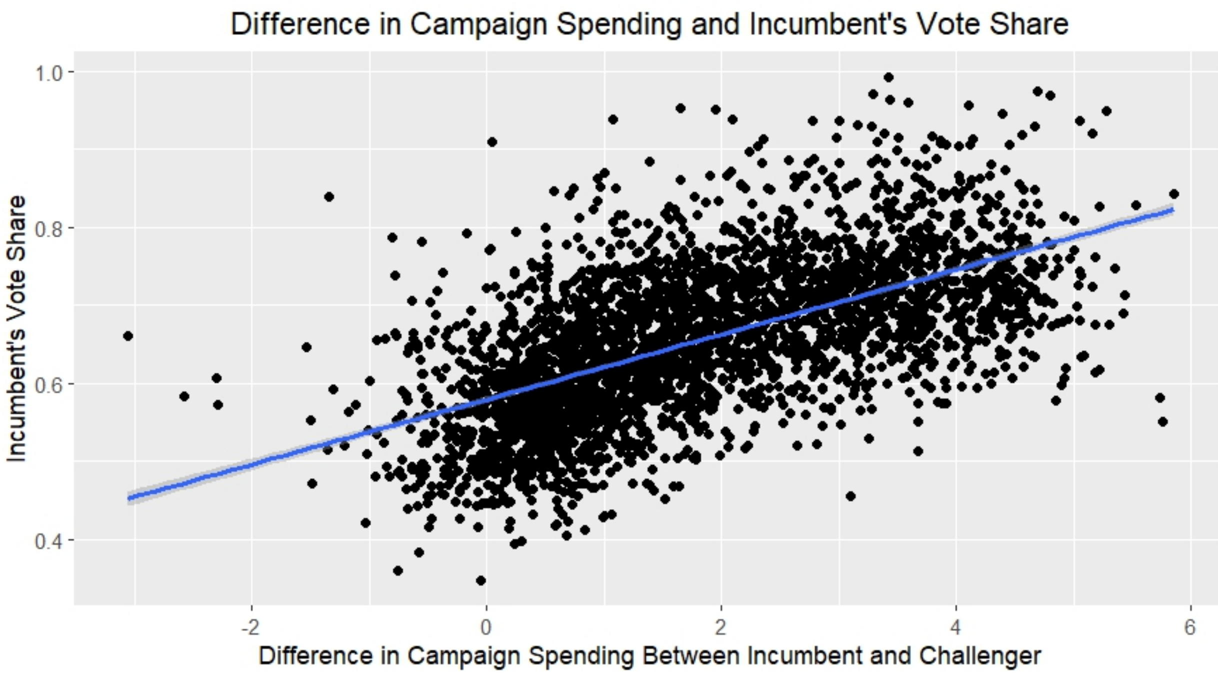
```
Call:
lm(formula = voteshare ~ difflog, data = inc.sub)

Coefficients:
(Intercept)      difflog
    0.57903      0.04167
```

2. Make a scatterplot of the two variables and add the regression line.

```r
theme_update(plot.title = element_text(hjust = 0.5))

ggplot(inc.sub, aes(difflog, voteshare)) +
  geom_point() +
  labs(title = "Difference in Campaign Spending and Incumbent's Vote Share",
       x = "Difference in Campaign Spending Between Incumbent and Challenger",
       y = "Incumbent's Vote Share"
  ) +
  geom_smooth(method = lm)
```



3. Save the residuals of the model in a separate object.

```r
regression1 <- lm(voteshare ~ difflog, inc.sub)
residuals1 <- regression1$residuals
```

4. Write the prediction equation.

   The numbers representing the slope and y-intercept are taken from the linear regression run in R.

$$\hat{y} = 0.04167x + 0.57903$$

   In the equation, y-hat represents the predicted value of the incumbent's vote share, and x represents the difference in campaign spending between incumbent and challenger.

Question 2

We are interested in knowing how the difference between incumbent and challenger's spending and the vote share of the presidential candidate of the incumbent's party are related.

1. Run a regression where the outcome variable is `presvote` and the explanatory variable is `difflog`.

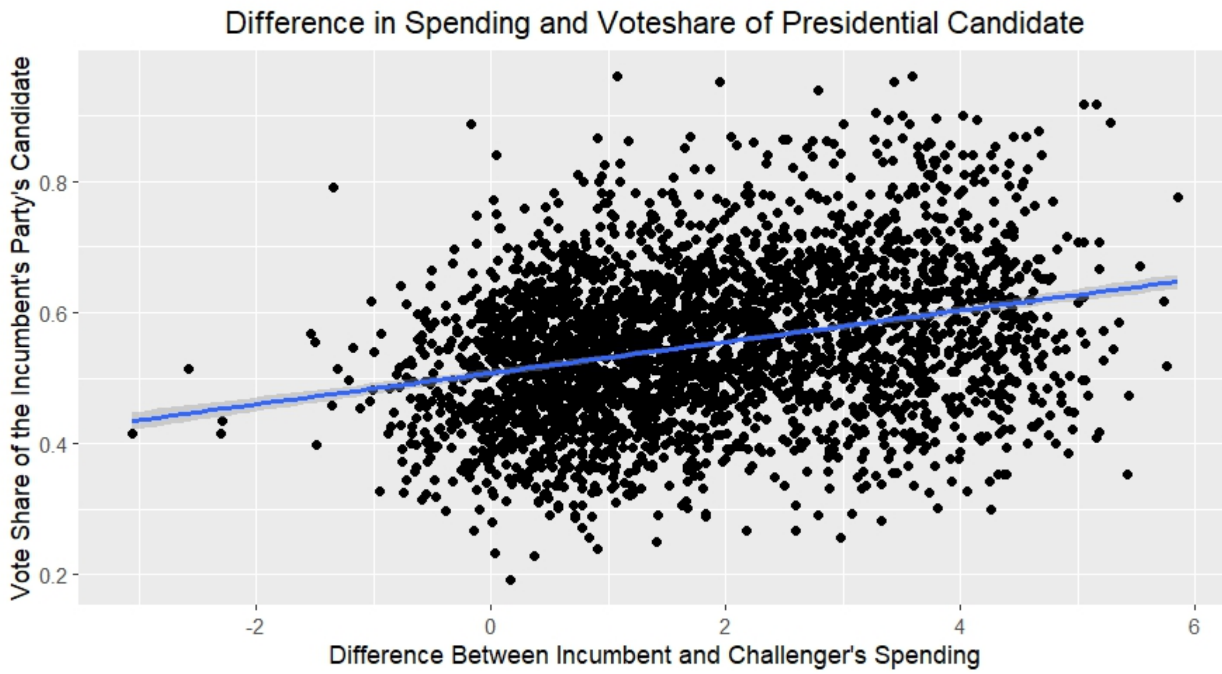
```
lm(presvote ~ difflog, inc.sub)
```

```
Call:
lm(formula = presvote ~ difflog, data = inc.sub)

Coefficients:
(Intercept)      difflog
    0.50758      0.02384
```

2. Make a scatterplot of the two variables and add the regression line.

```
ggplot(inc.sub, aes(difflog, presvote)) +
  geom_point() +
  labs(title = "Difference in Spending and Voteshare of Presidential
    Candidate",
       x = "Difference Between Incumbent and Challenger's Spending",
       y = "Vote Share of the Incumbent's Party's Candidate"
       ) +
  geom_smooth(method = lm)
```



3. Save the residuals of the model in a separate object.

```
regression2 <- lm(presvote ~ difflog, inc.sub)
residuals2 <- regression2$residuals
```

4. Write the prediction equation.

   The numbers representing the slope and y-intercept are taken from the linear regression run in R.

$$\hat{y} = .023837x + 0.507583$$

   In the equation, y-hat represents the predicted value of vote share of the presidential candidate of the incumbent party, and x represents the difference between incumbent's and challenger's spending.

Question 3

We are interested in knowing how the vote share of the presidential candidate of the incumbent's party is associated with the incumbent's electoral success.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `presvote`.

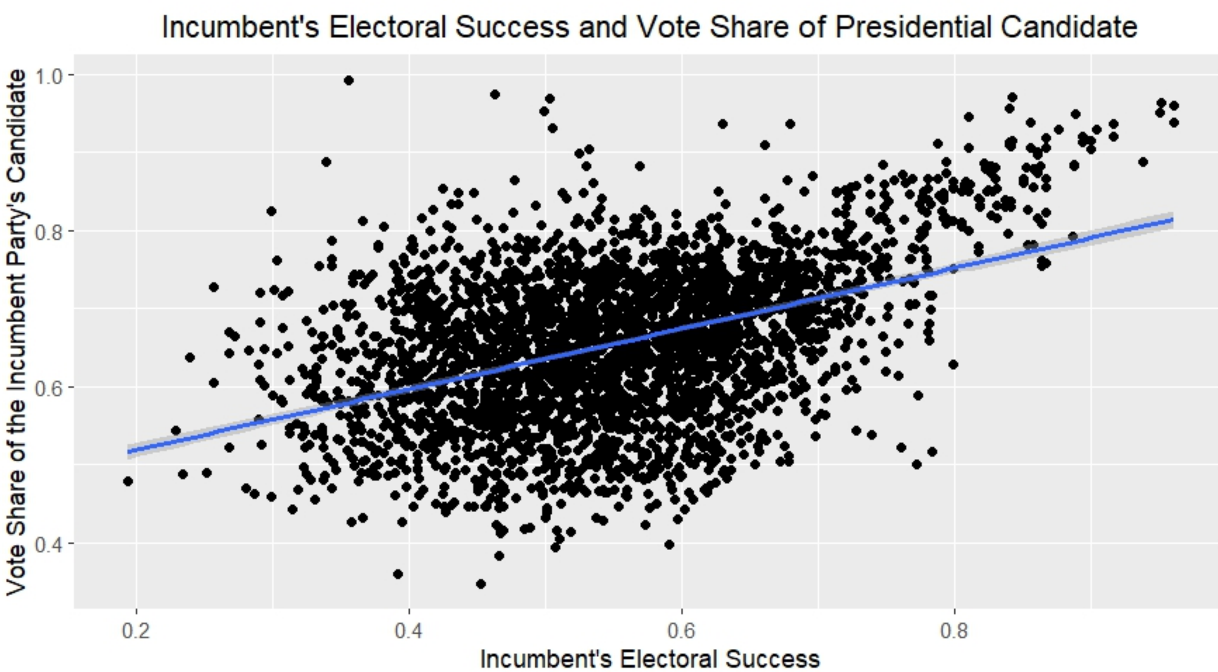
```
lm(voteshare ~ presvote, inc.sub)
```

```
Call:
lm(formula = voteshare ~ presvote, data = inc.sub)

Coefficients:
(Intercept)     presvote
     0.4413       0.3880
```

2. Make a scatterplot of the two variables and add the regression line.

```r
ggplot(inc.sub, aes(presvote, voteshare)) +
  geom_point() +
  labs(title = "Incumbent's Electoral Success and Vote Share of
    Presidential Candidate",
       x = "Incumbent's Electoral Success",
       y = "Vote Share of the Incumbent Party's Candidate"
  ) +
  geom_smooth(method = lm)
```



3. Write the prediction equation.

   The numbers representing the slope and y-intercept are taken from the linear regression run in R.

$$\hat{y} = 0.3880x + 0.4413$$

   In the equation, y-hat represents the predicted value of the incumbent's electoral success, and x represents the vote share of the presidential candidate of the incumbent's party.

Question 4

The residuals from part (a) tell us how much of the variation in `voteshare` is *not* explained by the difference in spending between incumbent and challenger. The residuals in part (b) tell us how much of the variation in `presvote` is *not* explained by the difference in spending between incumbent and challenger in the district.

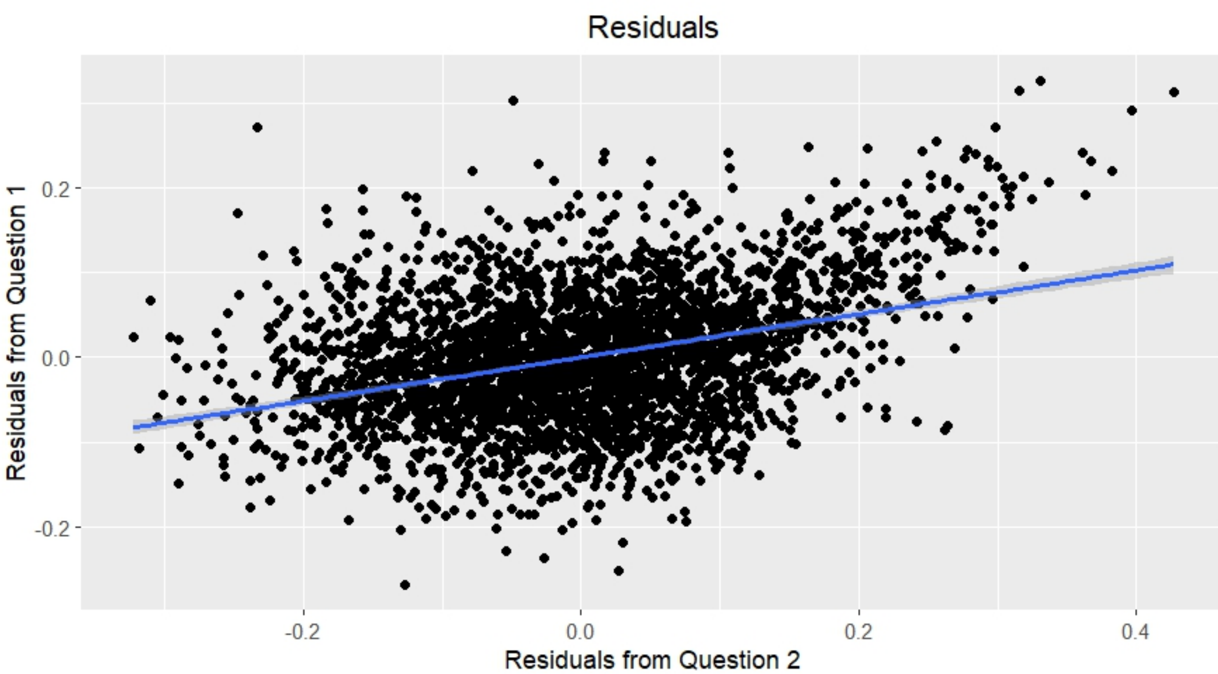
1. Run a regression where the outcome variable is the residuals from Question 1 and the explanatory variable is the residuals from Question 2.

```
lm(residuals1 ~ residuals2, inc.sub)
```

```
Call:
lm(formula = residuals1 ~ residuals2, data = inc.sub)

Coefficients:
(Intercept)                 residuals2
-0.000000000000000005934    0.256877012700097440145
```

2. Make a scatterplot of the two residuals and add the regression line.



3. Write the prediction equation.

   y = 0.256877012700097440145x - 0.000000000000000005934

Question 5

What if the incumbent's vote share is affected by both the president's popularity and the difference in spending between incumbent and challenger?

1. Run a regression where the outcome variable is the incumbent's `voteshare` and the explanatory variables are `difflog` and `presvote`.

```
lm(voteshare ~ difflog + presvote, inc.sub)
```

```
Call:
lm(formula = voteshare ~ difflog + presvote, data = inc.sub)

Coefficients:
(Intercept)      difflog     presvote
    0.44864      0.03554      0.25688
```

2. Write the prediction equation.

   The numbers representing the slope and y-intercept are taken from the linear regression run in R.

$$\hat{y} = 0.44864 + 0.03554x1 + 0.25688x2$$

   In the prediction equation, y-hat represents the predicted value of the incumbent's vote share, x1 represents the difference in spending between incumbent and challenger, and x2 represents the president's popularity.

3. What is it in this output that is identical to the output in Question 4? Why do you think this is the case?